EXPERIMENT – 02

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Aim

To install and configure Docker, pull Docker images, run containers, and manage the container lifecycle using Docker commands.

Objectives

- To pull Docker images from Docker Hub
- To run containers with port mapping
- To verify running containers
- To manage container lifecycle (start, stop, remove)

Theor y

Docker is an open-source containerization platform that allows applications to be packaged along with their dependencies into lightweight and portable containers. Containers run on a shared operating system kernel, making them faster and more resource-efficient than traditional Virtual Machines.

A Docker Image is a read-only template used to create containers.

A Docker Container is a running instance of a Docker image.

Docker follows a client–server architecture, where:

- The Docker Client sends commands
- The Docker Daemon executes them
- Docker Hub stores container images

Software Requirements

- Windows Operating System
- Docker Desktop (with WSL integration)
- Ubuntu (WSL distribution)

Procedure / Steps to Perform the Experiment

Step 1: Pull Docker Image

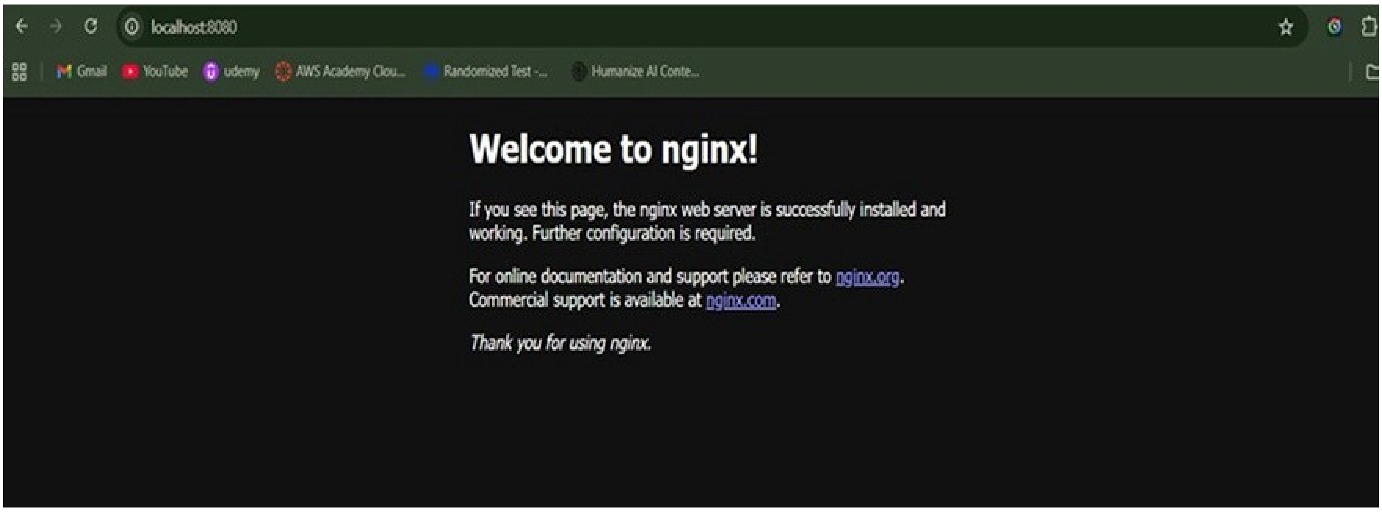
Pull the Nginx image from Docker Hub:

```
docker pull nginx
```

```
aryanraj@DESKTOP-DU3B1ES:/mnt/c/Users/Hp$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
46bf3a120c8e: Pull complete
bae5a1799a80: Pull complete
47cd406a84ef: Pull complete
0c8d55a45c0d: Pull complete
2e02dba24409: Download complete
a5d78d617315: Download complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Step 2: Run Container with Port Mapping

Run the Nginx container in detached mode:




```
docker run -d -p 8080:80 nginx
```

Explanation:

- `-d` → Runs container in background (detached mode)
- `-p 8080:80` → Maps host port 8080 to container port 80
- `nginx` → Docker image name

Step 3: Verify Running Containers

To check running containers:

```
docker ps
```

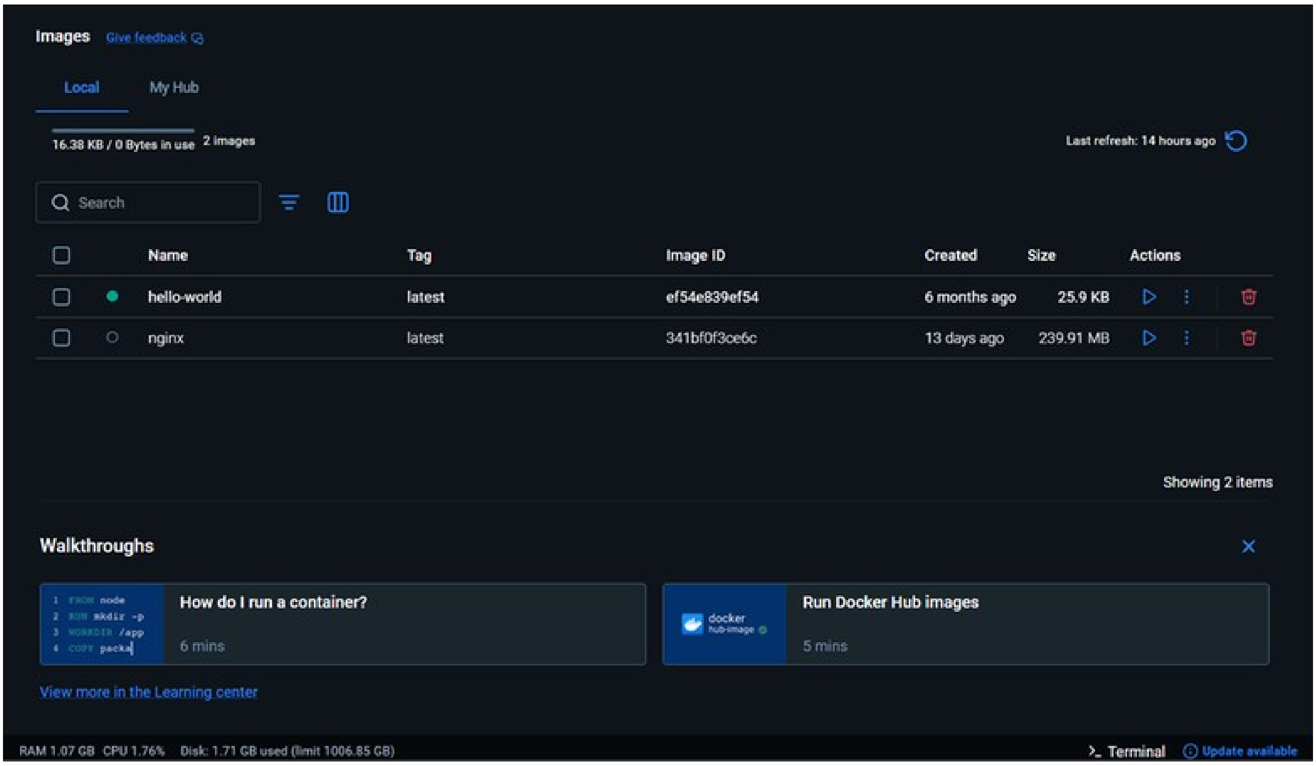
```
aryanraj@DESKTOP-DU3B1ES:/mnt/c/Users/Hp$ docker ps
CONTAINER ID   IMAGE          COMMAND     CREATED     STATUS     PORTS     NAMES
```

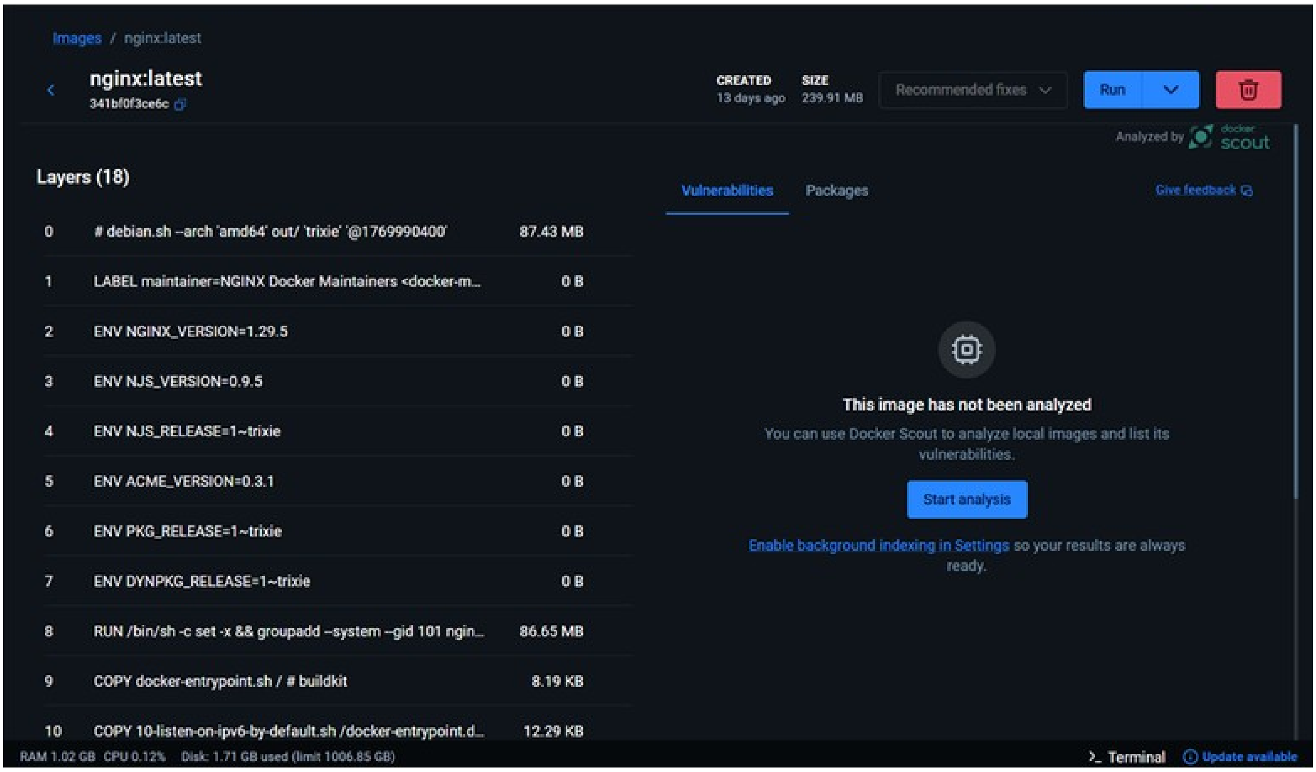
This command displays:

- Containe I
- Imag nam
- Container

Port mapping

```
aryanraj@DESKTOP-DU3B1ES:/mnt/c/Users/Hp$ docker stop nginx-container
```





Step 4: Stop and Remove Container

To stop the running container: `docker stop`

<container_id> To remove

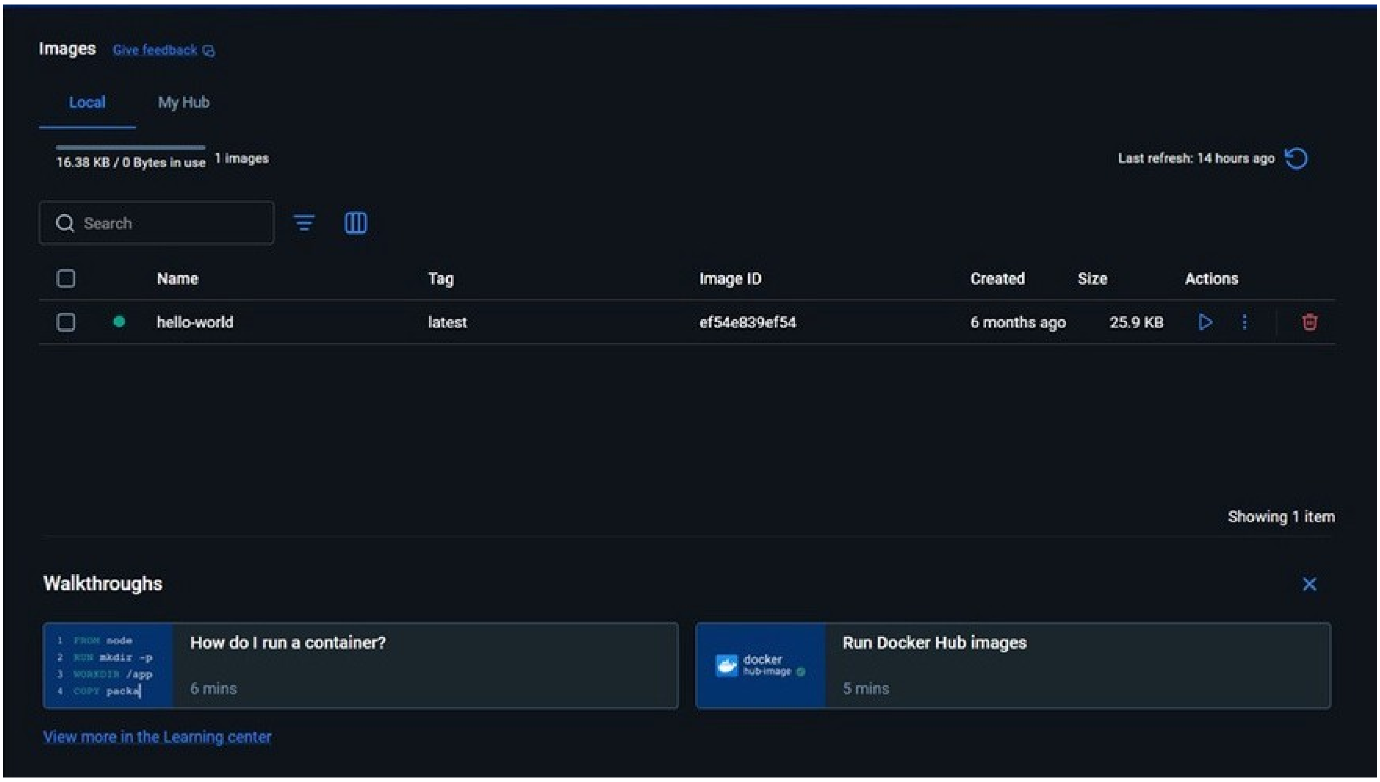
the container:

```
docker rm <container_id>
```

Step 5: Remove Docker Image

To remove the downloaded image:

```
docker rmi nginx
```



This frees disk space by deleting the unused image.

Result

Docker images were successfully pulled.
Containers were executed using port mapping.
Container lifecycle management commands (start, stop, remove) were performed successfully.